INTERNSHIP
PROGRAM
DATA ANALYTICS
45-DAYS
ApexPlanet
Software Pvt.Ltd
www.apexplanet.in

About Us

At ApexPlanet Software Pvt. Ltd., we are dedicated to driving innovation and excellence in the realm of digital solutions. Founded with a vision to empower businesses through cutting-edge technology, we specialize in delivering top-notch web development and app development services tailored to meet the unique needs of our clients.

Our team of skilled professionals is passionate about creating robust, scalable, and user-friendly applications that enhance user experiences and drive business growth. We leverage the latest technologies and industry best practices to ensure that our solutions are not only effective but also future-proof.

In addition to our core services, ApexPlanet Software Pvt. Ltd. is committed to nurturing the next generation of tech talent. We offer comprehensive internship programs that provide hands-on experience and mentorship, preparing aspiring developers for successful careers in the software industry.

Join us on our journey to make a significant impact in the digital world. With ApexPlanet Software Pvt. Ltd., your business is poised for success, and your career is set on a path of continuous learning and achievement.

How to Submit Your Internship Task

1. **Prepare Your Task**
   - Complete your task before the submission date.
2. **Create a Screen Recording**
   - Record your project using screen recording software and the video file.
3. **Upload Video to LinkedIn**
   - Log in to LinkedIn.
   - Go to your profile, add the video under "Featured."
   - Copy the video link.
4. **Upload Project to GitHub**
   - Create a GitHub account (if you don't have one).
   - Create a new repository and upload your project files.
   - Make the repository public and copy the link.
5. **Register and Log In to ApexPlanet**
   - Go to ApexPlanet Internship.
   - Register and log in.
4. **Submit Your Task**
   - Go to "Manage Task."
   - Verify using your offer letter ID and email.
   - Click on the relevant task button.
   - Fill in the details, paste the LinkedIn video link, and the GitHub repository link.
   - Submit.
5. **Open the Next Task**
   - After submitting, click to open the next task and repeat the process.

This streamlined process ensures your tasks are submitted professionally and on time.

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Data Analytics Internship
45 DAYS

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TASK - 1

TimeLine : 8 Days

Foundational Setup & Exploratory Data Analysis (EDA)

Objective:

- Set up analytics environment, understand data sources, perform exploratory data analysis, and document initial insights.

Activities:

1. **Day 1-2: Environment Setup**
   - Install Python (Anaconda distribution) and Jupyter Notebook / VS Code
   - Set up GitHub repository with structure:

```
apexplanet-data-analytics/
├── data/ (raw, processed)
├── notebooks/ (Jupyter notebooks)
├── scripts/ (Python scripts)
├── reports/ (PDF/PPT)
├── dashboards/ (PBIX/TWX files)
└── README.md
```

   - Install libraries: pandas, numpy, matplotlib, seaborn, plotly, scikit-learn, sqlalchemy
   - Create virtual environment (conda create -n analytics python=3.10)

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Foundational Setup & Exploratory Data Analysis (EDA)

2. **Day 3-4: Data Sourcing & Understanding**
   - Obtain dataset (choose one):
     - E-commerce sales data (Kaggle)
     - COVID-19 trends (Our World in Data)
     - Customer churn data (IBM Telco)
     - Supermarket sales (Tableau sample)
     - Or use provided company dataset
   - Load data into pandas DataFrame
   - Understand data dictionary (columns, types, descriptions)
   - Document data source, collection methods, and limitations
3. **Day 5-6: Data Cleaning & Preprocessing**
   - Handle missing values (drop, fill with mean/median/mode, or forward fill)
   - Remove duplicates
   - Fix data types (convert dates to datetime, categorical to category)
   - Handle outliers using IQR or Z-score method
   - Standardize/rename column names
   - Create cleaning log documenting all transformations

Foundational Setup & Exploratory Data Analysis (EDA)

4. **Day 7-8: Exploratory Data Analysis (EDA)**
   - Generate statistical summary (describe, info, value_counts)
   - Create univariate analysis: histograms, boxplots, bar charts
   - Create bivariate analysis: scatter plots, heatmaps (correlation matrix)
   - Identify patterns, trends, anomalies
   - Document key findings in a Jupyter notebook with markdown explanations
   - Save cleaned data to data/processed/

Deliverables

- GitHub: Jupyter notebook with full EDA + cleaned dataset + README
- LinkedIn post: "Task 1 complete at @Apexplanet! Environment setup, data cleaning, and EDA done on [dataset name]. Found [3 key insights]. #DataAnalytics #Python #EDA"

TASK - 2

TimeLine : 9 Days

SQL & Data Extraction

Objective:

- Master SQL queries for data extraction, transformation, and aggregation. Connect Python to databases.

Activities

1. **Day 9-11: SQL Fundamentals**
   - Set up local database (SQLite / PostgreSQL / MySQL using Docker or local install)
   - Import cleaned dataset into database
   - Practice SQL queries:
     - SELECT, WHERE, ORDER BY, LIMIT
     - JOIN (INNER, LEFT, RIGHT, FULL)
     - GROUP BY, HAVING, aggregate functions (COUNT, SUM, AVG, MIN, MAX)
     - Subqueries and CTEs (WITH clause)
     - Window functions (ROW_NUMBER, RANK, LAG, LEAD)
2. **Day 12-14: Advanced SQL**
   - Write complex queries for business problems:
     - Monthly sales trends
     - Top 10 customers by revenue
     - Customer retention rate
     - Product category performance
     - Moving averages and cumulative sums
   - Create views for frequently used queries
   - Optimize queries using EXPLAIN and indexing

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SQL & Data Extraction

3. **Day 15-17: Python + SQL Integration**
   - Connect Python to database using sqlalchemy and psycopg2/sqlite3
   - Execute SQL queries directly from Jupyter notebook
   - Use pandas.read_sql() to load query results into DataFrames
   - Create parameterized queries to avoid SQL injection
   - Build a reusable Python module (db_utils.py) for database connections

Deliverables

- GitHub: SQL query file (.sql) + Python scripts for DB connection + Jupyter notebook with SQL + Python integration
- LinkedIn post: "Task 2: SQL mastery complete! Wrote 20+ complex queries for business insights. Integrated Python with PostgreSQL for automated data extraction. #SQL #DataExtraction"

TASK - 3

TimeLine : 12 Days

Data Visualization & Dashboarding

Objective:

- Create professional dashboards using Power BI / Tableau and Python visualization libraries.

Activities

1. **Day 18-20: Python Visualization**
   - Create visualizations using Matplotlib:
     - Line plots (trends over time)
     - Bar charts (comparisons)
     - Histograms (distributions)
     - Scatter plots (correlations)
     - Subplots (multiple charts together)
   - Create advanced visualizations using Seaborn:
     - Heatmaps (correlation matrix)
     - Pairplots (multi-variable relationships)
     - Boxen plots (large datasets)
     - FacetGrid (multi-dimensional data)
   - Create interactive visualizations using Plotly:
     - Interactive line/bar/scatter plots
     - Dropdown filters and range sliders
     - Save as HTML for sharing

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TASK - 3

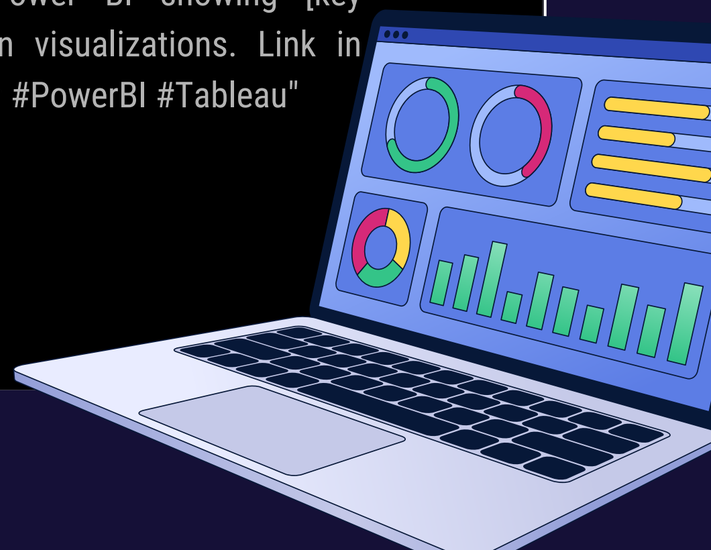
TimeLine : 12 Days

Data Visualization & Dashboarding

2. **Day 21-23: Power BI / Tableau Setup & Basics**
   - Install Power BI Desktop (free) or Tableau Public
   - Connect to cleaned dataset (CSV/Excel/database)
   - Learn interface: Fields, Visualizations, Filters, Pages
   - Create basic charts: bar, line, pie, map, table, matrix
   - Add slicers and filters
   - Create calculated columns and measures (DAX for Power BI / calculated fields for Tableau)
3. **Day 24-26: Advanced Dashboarding**
   - Build an executive dashboard with 6-8 visuals:
     - KPI cards (total revenue, customers, orders)
     - Time series trend (monthly sales)
     - Category breakdown (donut/bar chart)
     - Geographic map (sales by region)
     - Top 10 products/customers (bar chart)
     - Filter panel (date range, category, region)
   - Add drill-through pages for detailed analysis
   - Implement bookmarks and buttons for navigation
   - Apply company branding (colors, logo)
4. **Day 27: Dashboard Optimization & Publishing**
   - Optimize performance (reduce data load, use aggregations)
   - Add tooltips with secondary visualizations
   - Publish to Power BI Service / Tableau Public
   - Create a shareable link
   - Embed dashboard in GitHub README (if possible)

Deliverables

- GitHub: Python visualizations (HTML/PNG) + Power BI/Tableau file (.pbix/.twbx) + dashboard link
- LinkedIn post: "Task 3: Data visualization done! Built interactive dashboards in Power BI showing [key metrics]. Created 15+ Python visualizations. Link in comments. #DataVisualization #PowerBI #Tableau"



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TASK - 4

TimeLine : 11 Days

Advanced Analytics & Statistical Modeling

Objective:

- Apply statistical analysis, time series forecasting, customer segmentation, and basic predictive modeling.

Activities

1. **Day 28-30: Statistical Analysis**
   - Descriptive statistics (mean, median, mode, variance, standard deviation, skewness, kurtosis)
   - Inferential statistics:
     - Confidence intervals
     - Hypothesis testing (t-test, chi-square test, ANOVA)
     - p-values and significance levels
   - Correlation vs causation analysis
   - Use scipy.stats for statistical tests
   - Document findings with business implications

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TASK - 4

TimeLine : 11 Days

Advanced Analytics & Statistical Modeling

2. **Day 31-33: Time Series Analysis**
   - Convert data to time series format (set datetime index)
   - Resample data (daily, weekly, monthly)
   - Decompose time series: trend, seasonality, residuals (using statsmodels)
   - Check stationarity (ADF test)
   - Apply smoothing techniques: moving average, exponential smoothing
   - Build basic forecasting: ARIMA / SARIMA model
   - Evaluate forecast accuracy (MAE, RMSE, MAPE)
3. **Day 34-36: Customer Segmentation (Clustering)**
   - Prepare features for clustering (scale/normalize using StandardScaler)
   - Apply K-Means clustering
   - Find optimal K using elbow method and silhouette score
   - Interpret clusters (profile each segment by demographics, behavior)
   - Visualize clusters (PCA for 2D visualization)
   - Create segment-based recommendations

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TASK - 4

TimeLine : 11 Days

Advanced Analytics & Statistical Modeling

4. **Day 37-38: Predictive Modeling (Basic)**
   - Define target variable (churn, sales, etc.)
   - Split data into train/test sets
   - Build models:
     - Linear Regression (for numeric prediction)
     - Logistic Regression (for classification)
     - Decision Tree (for interpretability)
   - Evaluate models: R^2, MAE, RMSE (regression) / Accuracy, Precision, Recall, F1-Score, ROC-AUC (cla
   - Feature importance analysis
   - Document model limitations

Deliverables

- GitHub: Jupyter notebook with all statistical models + time series forecasts + clustering results + prediction models
- LinkedIn post: "Task 4 complete! Performed hypothesis testing, time series forecasting (ARIMA with 92% accuracy), customer segmentation (5 clusters), and churn prediction (85% recall). #DataScience #MachineLearning"

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TASK - 5

TimeLine : 7 Days

Final Reporting, Deployment & Presentation

Objective:

- Create executive summary report, automate analytics pipeline, record presentation, and submit final deliverables.

Activities

1. **Day 39-40: Executive Summary Report**

- Create professional PDF report (Google Docs / Canva / LaTeX) containing:
  - Executive summary (1 page)
  - Problem statement & objectives
  - Data source & methodology
  - Key findings with visuals (top 5-7 insights)
  - Statistical analysis results
  - Dashboard screenshots
  - Recommendations (actionable business suggestions)
  - Future scope & limitations
- Add table of contents and page numbers
- Use company branding (logo, colors)

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TASK - 5

TimeLine : 7 Days

Final Reporting, Deployment & Presentation

2. **Day 41-42: Automate Analytics Pipeline**
   - Create a Python script that:
     - Extracts data from source (CSV/database/API)
     - Cleans and transforms data
     - Runs analysis (calculates KPIs, refreshes models)
     - Exports results to CSV/Excel
     - Sends email report (optional using yagmail/smtplib)
   - Schedule script to run daily/weekly using:
     - Windows Task Scheduler OR
     - GitHub Actions OR
     - Apache Airflow (basic)
   - Document automation steps in README
3. **Day 43: Final Dashboard & GitHub Documentation**
   - Update dashboard with final insights
   - Publish dashboard to cloud (Power BI Service / Tableau Public)
   - Write comprehensive README.md:

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TASK - 5

TimeLine : 7 Days

Final Reporting, Deployment & Presentation

Apexplanet Data Analytics Project
Overview
- Objective- Dataset description
- Key questions answered
Tech Stack
- Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly
- SQL (PostgreSQL)
- Power BI / Tableau
- Scikit-learn, Statsmodels
Project Structure
- `/notebooks` - EDA, analysis, modeling
- `/scripts` - Automation scripts
- `/data` - Raw and processed data
- `/dashboards`
- PBIX/TWBX files

Key Findings
1. [Finding 1]
2. [Finding 2]
3. [Finding 3]
Dashboard Link
[Power BI / Tableau Public Link]
How to Run
1. Clone repo
2. Install requirements: `pip install -r requirements.txt`
3. Run Jupyter notebooks in order
Author
[Your Name]
Apexplanet Software Pvt. Ltd. Internship

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Final Reporting, Deployment & Presentation

4. **Day 44: Demo Video & Presentation**
   - Record 5-7 minute demo video:
     - Screen recording with voiceover (OBS Studio / Loom)
     - Script: Introduction → Problem statement → Data cleaning → EDA insights → SQL queries → Dashboard tour → Statistical analysis → ML model → Recommendations → Conclusion
   - Upload to YouTube (unlisted)
   - Create 8-10 slide presentation (Google Slides):
     - Slide 1: Title & Company
     - Slide 2: Problem & Objectives
     - Slide 3: Data Overview
     - Slide 4: EDA Insights (before/after cleaning)
     - Slide 5: Dashboard Screenshot
     - Slide 6: Statistical / ML Results
     - Slide 7: Business Recommendations
     - Slide 8: Challenges & Learnings
     - Slide 9: Future Scope
     - Slide 10: Thank You

TASK - 5

TimeLine : 7 Days

Final Reporting, Deployment & Presentation

5. **Day 45: Final Submission**
   - Clean GitHub repository (remove unused files, add requirements.txt, final commit tag v1.0.0)
   - Publish final LinkedIn post:

     45-Day Data Analytics Internship Completed at Apexplanet Software Pvt. Ltd.!*

     Transformed raw [dataset name] into actionable insights:

     Built interactive Power BI dashboard

     Performed statistical analysis & hypothesis testing

     Created time series forecast (92% accuracy)

     Customer segmentation (5 clusters)

     Churn prediction model (85% recall)

     GitHub: [link]

     Dashboard: [link]

     Demo: [YouTube link]

     Open to Data Analyst roles! #DataAnalytics #Python #SQL #PowerBI #InternshipComplete
   - Submit to Apexplanet: LinkedIn link + GitHub link

Deliverables

- GitHub: Final code + cleaned data + automation scripts + requirements.txt + README + final report PDF
- LinkedIn post (final submission with all links)

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Thank You for Your Attention

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